CS/IST 492 Intro. to Parallel Computing
Homework Assignment (3)

Objectives

- Understand MPI and OpenMP constructs
- Design and develop simple parallel applications for both message passing and shared-address space platforms.

Description

An approximation to the value of π can be obtained using the following expression:

$$\frac{\pi}{4} = \int_0^1 \frac{dx}{1+x^2} \approx \frac{1}{N}\sum_{i=1}^{N} \frac{1}{1+\left(\frac{i-\frac{1}{2}}{N}\right)^2}$$

Where the answer becomes more accurate as *N* is increased. The iterations over *i* are clearly independent and so the calculation can be done in parallel.

i. **(20 points)** Write a serial program to compute and print out an approximate value of PI. Test your code using different values of N. *Properly comment your code and provide screenshots of the output as well as the execution time.*

ii. Write two programs to solve this problem in parallel:
   a. **(25 points)** Use MPI to solve this problem and run on a distributed cluster. *Properly comment your code to explain your strategy of partitioning the iteration space among different processors.*
   b. **(25 points)** Use OpenMP to allow your program to run in a multithreaded fashion. *Properly comment your code to explain your solution and the rational behind using different constructs.*

iii. **(10 points)** Run your programs on *Scholar* and test them using N=100,000 and 10 processes. *Provide screenshots of the output as well as the execution time in each case.*

iv. **(20 points)** Compute and comment on the speedup using 8, 16, and 32 processes for each of your programs. *Provide screenshots of the outputs as well as the execution time in each case*.

*Submit **ONE** neatly formatted "pdf" file. Don't forget to number your answers accordingly.*